

Leveraging Game Theory for Intelligent and Adaptive Traffic Signal Systems

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Abstract- Traffic congestion is a major growing problem in urban areas, where inefficient signal timing plans result in delay, emissions, and safety risks. This paper proposes an adaptive traffic signal controller for an isolated intersection utilizing connected vehicle telemetry and game-theoretical principles. This approach resulted in a fair allocation of green time to each signal phase. The proposed model was built on Python and SUMO software, and a comparison was made against a predetermined fixed signal timing plan. The model was then evaluated using (Automated Traffic Performance Measures) ATSPMs and standard intersection measures. Assess the model on an isolated intersection in Columbus, Ohio, and the model showed a significant performance increase. Results showed an improvement of 29.62% in total delay, 28.69% in average speed, 5% in travel time, 15% average queue length, 19.83% in average fuel consumption, 19.83% in carbon monoxide emissions, 45.8% in split failures, and 4% in arrivals on red relative to the baseline model.

Keywords: game theory, adaptive signal control, delay, ATSPMs.

I. INTRODUCTION

Traffic congestion remains a persistent operational challenge at signalized intersections, leading to increased travel time, emissions, fuel consumption, and safety risks. The main challenge lies in the management of signalized intersections, which are control points of traffic throughput. Traditional traffic signal timing plans have been prepared using historical traffic data. In return, this has resulted in fixed traffic signal control (FTC), which operates on pre-defined cycle length and phase splits [1]. While such methods are straightforward to implement, they are not responsive to real-time traffic fluctuations. At the complex network level, this approach expands, requiring elements such as offsets, phase split constraints, cycle constraints, and queue spillback control. Hence, inadequate predetermined settings can lead to poor network progressions. Furthermore, FTC has been reported with bad vehicle progression at varied demand at both the isolated and network levels [2].

Traffic operators have then shifted to adaptive traffic signal controls (ASC), which operate with fluctuating vehicle demand. The common approaches are Split Cycle Offset Optimization Technique (SCOOT), Sydney Coordinated Adaptive Traffic System (SCATS), Real-Time Hierarchical Optimized Distributed Effective System (RHODES), Optimized Policies for Adaptive Control (OPAC), "Virtual Fixed Cycle", and ACS Lite [3], [4]. These models have been successfully employed in both centralized and decentralized logics. These models have been reported to demand regular parameter tuning for efficient performance since they are limited to phase reordering and oversaturation sensitivity. Most

importantly, they depend on detector data for calibration, which is expensive to maintain and install. In a scenario of insufficient data quality, advanced sensors are employed, increasing operational costs [5], [6].

The emergence of connected vehicles (CVs) over the past decade presents a whole new solution. CVs act as mobile sensors continuously transmitting rich, high-frequency data such as speed, position, acceleration profile, and route intentions [7], [8]. Unlike loop detectors or roadside sensors that provide localized and often delayed information, CV technology enables network-wide and real-time traffic awareness. This continuous data stream can greatly enhance ASC systems by reducing reliance on fixed detectors. This also enables decentralized and centralized control, with improved predictive capabilities of the ASC systems. Hence, this study finds that integrating CV data into ASC systems is rich as it addresses the common limitations in real-time traffic control.

Fair signal timings allocation is one of the critical decisions in traffic control management. Game theory (GT) provides the mathematical framework for fair decision-making in competing scenarios. Studies have successfully implemented several cooperative and non-cooperative games in unsignalized intersections where vehicles act as rational agents. [9]. Nevertheless, in signalized intersections, studies have reported several game models with several limitations. One of the major limitations reported in signalized settings is computational difficulty and convergence of equilibrium solutions in varied demand [10]. Hence, this study proposes a centralized greedy maximization strategy using CV telemetry, considering both motor vehicles and pedestrians.

In addition to standard metrics such as delay, average travel time, emissions, and queue length, we report Automated Traffic Signal Performance Measures (ATSPMs). ATSPMs allow traffic engineers to quickly and continuously measure the impact of signal timing changes, reducing the need to develop a model to represent traffic conditions [11]. The remainder of this paper is organized as follows. Section II reviews related works. Section III formulates the proposed methodological approach. In Section IV, we present the results. Section V presents the discussion of results. Section VI concludes the study with limitations and future research directions.

II. RELATED WORKS

The Highway Capacity Manual (HCM) approach to signal timings is the world-renowned baseline approach in the design of traffic signal timings. This study acknowledges the contribution of the Transportation Research Board in the

development of this manual [12]. Nevertheless, we identify several limitations of predetermined signal timing approaches when traffic demand fluctuates or oversaturates in real time. This call justifies the need for adaptive traffic signal control.

Studies have extensively explored other disciplines, such as computer science and economics, to provide novel adaptive traffic signal controllers. In computer science disciplines, such as reinforcement learning, deep learning, and other machine learning models have been adopted. Studies have reported promising improvement in terms of average delay and travel time under simulation runs [13], [14], [15]. One of the limitations of these models is their real-world applicability. The models require an extensive training process on high-quality data, making them operationally expensive. Also, this goes along with the need for high computational power, demanding fast computing devices to increase operational costs. Nevertheless, the unexplainable black box in these models has raised concerns about their use in real-world scenarios.

In light of these limitations, GT has been proposed in recent years as an optimization decision-making tool for signalized and unsignalized intersections. Equilibrium solutions such as the Nash bargaining solution and the Stackelberg solution have been employed to increase vehicle users' comfort, safety, and mobility in unsignalized intersections [16], [17]. While in signalized intersections, they have been used to decrease delay, queue lengths, and emissions [18], [19], [20]. In both cases, single or multi-objective models have been proposed to optimize the intersection's performance. Focusing on signalized intersections, reports show a significant improvement in traffic congestion compared to the baseline models. Although one of the significant challenges is the development of GT models in dynamic traffic environments. An intersection can exhibit marked changes in terms of the arrival of vehicles, road conditions, or vehicle states. Hence, the proposed models don't account for the dynamic traffic conditions. Also, the proposed GT solutions have been reported to be computationally complex, making them challenging to implement in real-time traffic management systems [10], [21]. This calls for efficient and lightweight algorithms that can still perform under the same GT principles. Looking at the performance measures in these studies, there is also a need to reflect on what the real world is using, and that is ATSPMs. ATSPMs have been the leading key indicators of traffic signal performance. Studies have not reported ATSPMs such as arrivals on green, arrivals on red, split failures, green occupancy ratios, or red occupancy ratios, but rather used standard measures such as delay and travel time. ATSPMs, having been more detailed measures, studies are urged to present these metrics for easy understanding by traffic engineers. Hence, we find it necessary to discuss several ATSPMs.

Effective means of communication between vehicle-to-vehicle or V2I have been used as an effective means to help model the proposed above approaches. Studies have shown that with a modest penetration rate of CVs, significant improvements in traffic signal performance can be achieved [2], [4], [7], [8]. The integration of CV data can facilitate proactive decision-making, leading to reduced delays and

improved intersection throughput. Despite the advancements in adaptive and game-theoretic traffic signal control, challenges remain in terms of computational complexity and real-time applicability. Many existing models require extensive computational resources, limiting their practicality in real-world settings. To address these issues, this study proposes a lightweight, utility-driven game-theoretic controller tailored for CV environments. By leveraging CV telemetry and GT principles, this study presents an adaptive traffic signal controller.

III. METHODOLOGY

A. Fixed time signal settings

This study focused on an isolated intersection with a typical phase sequence as shown in Fig. 1.

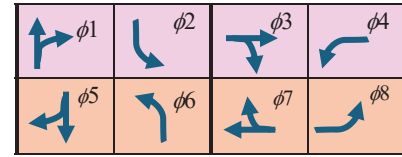


Fig. 1. (a) Signal Phase Sequence

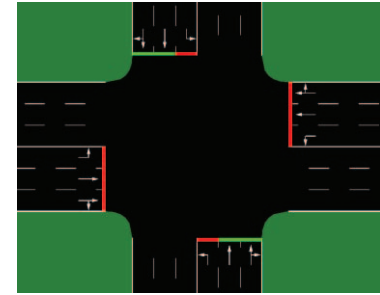


Fig 1. (b) Standard Intersection Layout

After careful estimation of critical flow rates, V_c , we follow the HCM approach [12] for FTC to determine the green signal timings for each phase. Table 1 presents a typical signal plan.

TABLE I. HCM results

Symbol	Phase	V_c (veh/h/ln)	Split	Split time (s)	Loss (Y+R) (s)	G (s)
$\phi1, \phi5$	NS	357	0.380	46	4	42
$\phi3, \phi7$	EW	362	0.386	46	4	42
$\phi2, \phi6$	NS_LT	112	0.119	14	4	10
$\phi4, \phi8$	EW_LT	108	0.115	14	4	10
	<i>sum</i>	939	1.000	120	16	104

B. Problem formulation

Consider an isolated four leg intersection with protected left turns and through phases as shown in Fig 1. (b) This configuration results into 4 players who compete for green time in the same cycle sequence. Through phases ($\phi1, \phi5$) denoted as NS, compete with phases ($\phi3, \phi7$) denoted as EW in the same game. While left turn phases ($\phi2, \phi6$) are denoted as NS_LT, compete with phases ($\phi4, \phi8$) denoted EW_LT for left turn green

time. This results in 2 non-cooperative games where each player in each game seeks an optimal green time solution.

$$\text{Players, } P: P_1 = \{NS, EW\}, P_2 = \{NS_{LT}, EW_{LT}\} \quad (1)$$

At any time one player is served with green time and the opposing player and is stopped, the objective is to reduce delay and queue length to the phase with the greatest immediate benefit while considering the maximum possible green time. Hence, the possible green time that can be allocated to each player is provided as:

$$g_{P_i}(t) = [g_{base}, g_{max}] \quad (2)$$

Where t is the decision epoch in a cycle, and g_{P_i} is the allocated green time. We arrive at this decision with a weighted objective function stating:

$$U_{P_i}^b(g_i, g_{-i}) = -(\alpha \cdot Q_{P_i} + \beta \cdot W_{P_i}), \quad \text{for: } \alpha = 1 - \beta \quad (3)$$

Where:

$$Q_{P_i}(t) = \sum_{l=L_{P_i}} q_l(t), \quad W_{P_i}(t) = \sum_{l=L_{P_i}} w_l(t) \quad (4)$$

Where α , and β weights assigned to queue length and waiting time in the utility function before assigning green time, $U_{P_i}^b$. After assigning the green time, we re-evaluate the utility value, U_i^a With the allocated green time as:

$$U_i^a(g_{P_i}, g_{-i}) = -(\alpha \cdot Q_{P_i} + \beta \cdot W_{P_i} - \gamma \cdot g_{P_i}) \quad (5)$$

The allocated green time abides to the following constraints:

$$g_{P_i} = \min(g_{base} + \gamma \times W_{P_i}, g_{P_i, max}) \quad (6)$$

$$\sum_{i=1}^2 g_{P_i} \leq 2(Y + g_{P, max}) \quad (7)$$

C. Solution approach

To find the optimal solution strategy, this study adopts a greedy utility maximization strategy. Unlike classical equilibrium-based models, this strategy selects optimal green times at each decision cycle to minimize overall vehicle waiting time and length of queue, by allowing incremental increase of green time to the phase. The greedy utility strategy iterates over all possible actions it can take and calculates the immediate utility. In return, in each decision cycle, the controller evaluates all feasible green time allocations (g_{P_i}) and sets the proposed green time. The solution can be presented as:

$$g'_{P_{1i}}(t), g'_{P_{2i}}(t) = \arg \max_{p \in P} [U_{P_{1i}}(t) + U_{P_{2i}}(t)] \quad (8)$$

D. Ablation Study

We then conducted an ablation study using SUMO software on the weights placed on queue length, waiting time, and green extension. To our findings, smaller incremental values of γ ranging from 0.03 to 0.01 allowed fast convergence of the model, better performance in terms of delay (19%-52%), travel time (3.84%-8.86%), and mean speed (56%-60%). Also, these values restored the adaptive nature of the model at all times while offering fair green times to congested approaches in real time. When, γ was held constant at its optimal solution, we varied the α , and β alternating from [0,1] to find that there was

minimal percentage improvement at higher values of waiting time ($\beta \geq 0.6$) +0.2% percentage improvement of the model resulting from left turn lanes. A further study was done to identify the relationship between waiting time and queue length. Fig 2 (b) shows this relationship. We discovered a linear relationship between queue length and waiting time in through lanes, explaining the model irresponsible behavior towards improvement of the weights in through phases. Also, a constant linear relationship was observed in left turn lanes, where at a constant queue length, there was a significant increase in waiting time. This justifies our earlier discussion that more weight on waiting time leads the model to an optimal solution in left turn lanes, as they are bound to shorter queue lengths due to traffic demand compared to through lanes.

E. Study area and Simulation settings.

Our key area of focus in this paper is an isolated intersection. Our study experiments 2 study areas. Our first study area is a standard 4 leg isolated intersection built with synthetic traffic data to explain our model's overall behavior. Each approach has 3 lanes having 1 left turn lane, 1 through lane, 1 through and right turn lane. The proposed vehicle data is found in Table I. All lanes had a speed limit of 15m/s (33.5mph). To create a mixed traffic flow, we allocated varied vehicle demand simulated at peak and off-peak sessions using 3 types of vehicles: passenger cars, buses, and single-unit trucks with varied vehicle performances.

Our second study area is an intersection joining Columbus Pike Road & East Powell Road in Columbus, Ohio. We used vehicle turning movement volumes recorded on Wednesday, 1st May 2024, to evaluate our model's performance during evening peak hours. Vehicle data was accessed from the Ohio Department of Transportation (ODOT) website [22]. After cleaning all the edges and unwanted polygons, Fig. 2 (a) presents the study area of focus. After cleaning the data, at peak hour, the total critical volume was found to be 2095 veh/hr/ln, where 912, 737, 252, and 194 veh/hr/ln were in NS, EW, NS_LT, and EW_LT phases, respectively. Following the HCM approach, 55s, 43s, 12s, and 9s were the effective green times assigned for NS, EW, NS_LT, and EW_LT phases, respectively.

F. Implementation

The simulation was developed and implemented in the SUMO simulator using Python integrated with the traffic control interface (TraCI) module. We evaluated the model under various random seeds, i.e., 10, 20, 30, 42, 50, 64, and 84. We found out that the results were highly consistent with improvements in travel time by $\pm 0.2\%$, mean speed by $\pm 1.3\%$ and delay by $\pm 0.4\%$ at a 95% confidence interval.

The controller executes continuously for each arriving vehicle in the intersection and adapts to the time-varying demand, providing fairness and efficiency without requiring centralized optimization.

TABLE II. Summary of Results – Standard Intersection

Controller/Metrics	Avg. Travel time (s)	Mean Speed (m/s)	Avg. Queue Length (veh)	Avg. Fuel Cons. (mg/s)	AoG (%)	AoR (%)	Total Delay (s)	CO Emiss. (mg/s)	Total Split failures
GTC	63.47	7.54	103.3	46,732	58.0	42.0	313.9	3784.7	4
FTC	69.45	4.77	127.7	48,945	65.7	34.3	455.4	4258.4	1
Improvement (%)	9.43	57.81	19.1	4.52	-	-	31.05	11.12	-

Table III. Summary of Results – Columbus Intersection

Controller/Metrics	Avg. Travel time (s)	Mean Speed (m/s)	Avg. Queue Length (veh)	Avg. Fuel Cons. (mg/s)	AoG (%)	AoR (%)	Total Delay (s)	CO Emiss. (mg/s)	Total Split failures
GTC	94.91	5.42	355.1	119,554	72.5	27.5	1609.9	10,618	52
FTC	100.03	4.21	421.6	107,011	69.7	30.3	2277.5	13,244	96
Improvement (%)	5.12	28.69	15.5	19.83	4.02	4.02	29.62	19.83	45.8

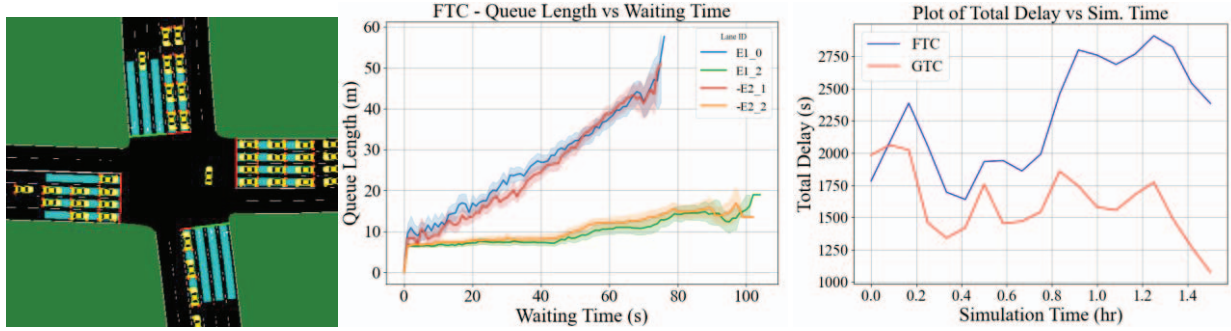


Fig. 2. (a), (b), (c) - (From Left to Right): 2(a) Columbus Intersection Layout from SUMO-gui. Fig 2(b) Plot of Queue Length versus waiting time for left turn lanes: E1_2 and -E2_2, and through lanes: E1_0 and -E2_1. Fig 2(c) Plot of Total delay versus simulation time.

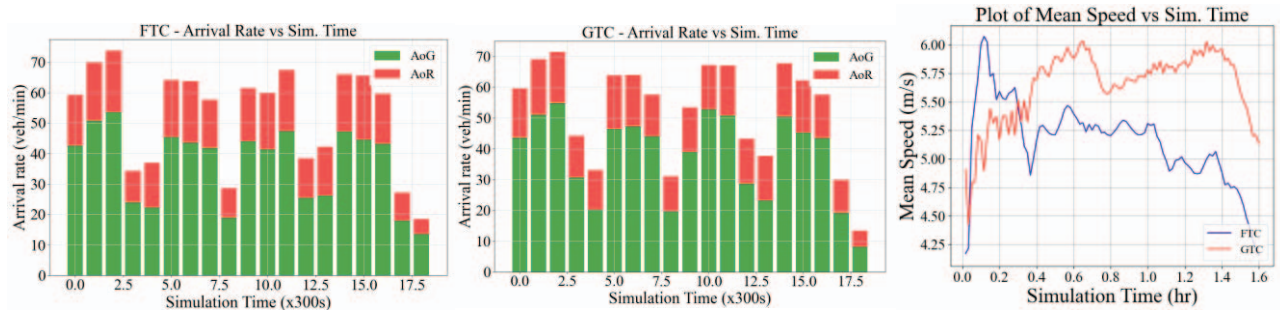


Fig. 3. (a), (b), (c) - (From Left to Right): Fig 3(a) FTC Arrival Rate versus simulation time. Results are aggregated in a 5-minute cycle. 3(b) GTC Arrival Rate versus simulation time. Fig 3(c) Plot of mean speed versus simulation time.

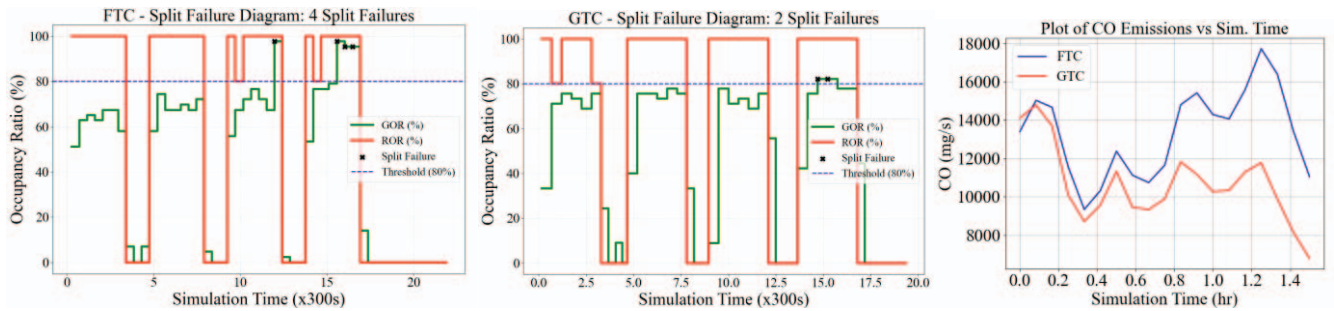


Fig. 4. (a), (b), (c) - (From Left to Right): 4(a) FTC Split Failure Diagram for Eastbound and Westbound Through Phase 4(b) GTC Split Failure Diagram for Eastbound and Westbound Through Phase. Fig 4(c) Plot of Carbon monoxide (CO) emissions versus simulation time.

NOTE: GOR - Green Occupancy Ratio, ROR - Red Occupancy Ratio.

G. Performance metrics

In addition to average travel time, delay, average speed, fuel consumption, and emission metrics, we introduce ATSPMs. The ATSPMs reported are Split failures, Arrival on Green (AOG), and Arrivals on Red (AOR). A split failure event indicates that a signal cycle does not provide enough green time to serve vehicles waiting for the phase. It occurs when the green occupancy ratio (GOR) and red occupancy ratios (ROR) are above 80%. On the other hand, AOG measures the arrivals of vehicles during the green phase, while AOR indicates the percentage of vehicles arriving during the red phase. A good progression is indicated with a higher AOG (60-80%). $AOG(\%) = 100\% - AOR(\%)$. "GTC" is used to present the proposed Game Theoretic Controller.

IV. EXPERIMENTAL RESULTS

The performance of the proposed GTC was evaluated against the conventional FTC using standard intersection and then using real-world data. We limit our study to the FTC, given that it is the underscored control that is adopted worldwide, unlike other proposed adaptive traffic controllers. We acknowledge the contributions of researchers in deploying several adaptive control models, although in this study, we discuss FTC in comparison to GTC.

On a standard intersection, the simulation was run for a 4-hour duration on both peak and off-peak hours. Table 2 provides a summary of all the performance measures. The results show an increased performance in average travel time (+9.73%), mean speed (+57.81%), average queue length (+19.1%), average fuel consumption (+4.52%), total delay (+31.05%), and carbon monoxide emissions (+11.12%). This showed a significant improvement in intersection performance under GTC. In terms of ATSPMs, GTC performed less compared to FTC in both AOG, AOR, and total split failures. We revise these trade-offs with real-world demand below.

When evaluated on field-calibrated demand with an evening peak from 4:45 pm to 5:30 pm, GTC outperformed STC in all measures. Table 3 provides a summary of all the performance measures. We observed an increased performance in average travel time (+5.12%), mean speed (+28.69%), average queue length (+15.5%), average fuel consumption (+19.83%), total delay (+29.62%), and CO emissions (+19.83%). In terms of ATSPMs, 4.02% increase in AOR/AOG and in split failures by 45.8% reflecting more equitable and timely green allocations across phases. These results are supported by fig. 2(c), fig 3 (a)-(c), and fig 4(a)-(c). Fig. 2(c) shows how the total system's delay was decreased against simulation time by 29.62%. We plot a compound bar graph for AOR and AOG in Fig. 3(a) and (b). Data collected during the simulation runs were aggregated to 300s, and AOR and AOG were estimated. We aggregate these results for ease of visualization as seen in Fig. 3(a) and (b). Similarly, Fig. 4(a) and (b) represent typical split failure diagrams for Eastbound and Westbound Through phases for both FTC and GTC, respectively. Split failures occur for each phase and are measured on each stop bar for each lane. Hence, a total of 8 split failure diagrams are generated for both FTC and GTC, detailing the occurrence of split failures in each

phase. Due to the space limitations, we present 2 of 8 diagrams. GOR presents the Green occupancy ratio, while ROR presents the red occupancy ratio. These reductions are attributed to smoother acceleration profiles, minimized idling times at the intersection and fewer harsh stops.

Overall performance implies that GTC was more effective in managing queues and lowering delays compared to FTC. This prevents excessive accumulation of congestion. Fuel consumption and pollutant emissions are critical concerns in traffic operations. Both pollutants and fuel consumption have been lowered, indicating better environmental impact and an economic approach to saving fuel.

V. DISCUSSION

The significant reduction in waiting time and delay highlights the core advantage of the adaptive traffic signal controller using the GT approach over STC. By evaluating utility functions for each phase, GTC dynamically adjusted green splits to favor more congested movements while preventing the over-penalization of minor approaches. The greedy maximization strategy proved to be effective as the algorithm easily converged at each decision step, delivering an optimal solution to the signal timings.

Furthermore, the improved mean speed and shortened queue lengths provide an initial indication of the GTC performance at a network level. Smoother progression reduces not only individual vehicle delays but also secondary congestion effects such as backward queue propagation. The environmental gains observed, such as lower fuel consumption and emissions, are consequences of fewer acceleration-deceleration events and reduced idling time at intersections. This supports the broader sustainability objectives of modern intelligent transportation systems. The ability to achieve both operational efficiency and environmental improvements positions the GTC as a critical aspect for future smart city implementations where energy savings and emission reductions are increasingly prioritized.

VI. CONCLUSION

This study proposed an adaptive traffic signal control framework modeled by game theoretical principles, optimizing traffic signal operation using connected vehicles (CVs) telemetry. By modeling competing traffic movements as rational players seeking to maximize utility, the controller dynamically allocated green time based on queue length and waiting time, achieving real-time responsiveness without relying on complex optimization algorithms. Experiments in SUMO using TraCI, and evaluated through both ATSPM and microsimulation metrics, proved that the proposed model is effective compared to FTC. The proposed GTC consistently reduced total delays, shortened queue lengths, and improved overall travel time and mean speed. Despite a decrease in arrivals on the green, the controller demonstrated improved operational efficiency and fairness, validating the effectiveness of utility-based green time allocation.

These findings underscore the potential of game theory-inspired methods, such as low-complexity and high-impact

solutions for intelligent traffic control in CV environments. The interpretability and modularity of the approach make it suitable for integration into urban intersections with existing infrastructure and scalable to multi-intersection networks. This model can be used in a connected vehicle environment where there is efficient CV communication to the traffic controller. This study is limited to assuming the vehicle data only contains CV. Future research will consider a mixed traffic environment to underscore the performance of the model outputs.

The proposed model can be scaled to a network level where a parent game can be built to control intersection signal coordination. In setting the game in a multi-intersection scenario, more ATSPMs can be considered to further explain the model performance. Future research work will focus on extending the model to support multi-intersection systems and incorporating advanced game-theoretical approaches. Additionally, future work will explore hybrid approaches that combine CV data with traditional sensor inputs to ensure robustness under varying technological adoption rates. While simulation reflects real traffic conditions through varied flow and vehicle behavior, this study is limited to simulation testing only. Field testing in a live CV environment can be pursued to validate the simulation findings and evaluate practical deployment feasibility. Nevertheless, we tested the model on real-world data, and we noticed a significant performance increase. This indicates promising results on using the greedy maximization strategy.

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REFERENCES

- [1] M. Eom and B. I. Kim, "The traffic signal control problem for intersections: a review," Dec. 01, 2020, *Springer Science and Business Media Deutschland GmbH*. doi: 10.1186/s12544-020-00440-8.
- [2] J. Li *et al.*, "Adaptive signal control and coordination for urban traffic control in a connected vehicle environment: A review," *Digital Transportation and Safety*, vol. 2, no. 2, pp. 89–111, 2023, doi: 10.48130/dts-2023-0008.
- [3] FHWA, "Adaptive Signal Control Technology." Accessed: Apr. 23, 2025. [Online]. Available: <https://www.fhwa.dot.gov/innovation/everydaycounts/edc-1/asct.cfm>
- [4] J. Guo, L. Cheng, and S. Wang, "CoTV: Cooperative Control for Traffic Light Signals and Connected Autonomous Vehicles using Deep Reinforcement Learning," Jan. 2022, [Online]. Available: <http://arxiv.org/abs/2201.13143>
- [5] P. Mirchandani and L. Head, "A real-time traffic signal control system: architecture, algorithms, and analysis q," Jun. 2000. [Online]. Available: www.elsevier.com/locate/trc

- [6] P. Jing, H. Huang, and L. Chen, "An adaptive traffic signal control in a connected vehicle environment: A systematic review," Aug. 24, 2017, *MDPI AG*. doi: 10.3390/info8030101.
- [7] J. Wang, S. Jiang, Y. Qiu, Y. Zhang, J. Ying, and Y. Du, "Traffic Signal Optimization under Connected-Vehicle Environment: An Overview," *J Adv Transp*, vol. 2021, 2021, doi: 10.1155/2021/3584569.
- [8] Z. Mo, W. Li, Y. Fu, K. Ruan, and X. Di, "CVLight: Decentralized learning for adaptive traffic signal control with connected vehicles," *Transp Res Part C Emerg Technol*, vol. 141, Aug. 2022, doi: 10.1016/j.trc.2022.103728.
- [9] Z. Wu, W. Xue, J. Huang, and Y. Chen, "A Differential Game Approach for Vehicle Competition and Cooperation at Unsignalized Intersections," in *2022 International Conference on Advanced Robotics and Mechatronics (ICARM)*, 2022, pp. 533–538. doi: 10.1109/ICARM54641.2022.9959115.
- [10] H. Abdelghaffar, H. Yang, and H. Rakha, "Isolated Traffic Signal using Nash Bargaining Optimization," *Global Journal of Research in Engineering*, vol. XVI, 2016.
- [11] A. Tanaka, B. Schroeder, L. Trask, and T. Chase, "NCDOT GUIDE ON AUTOMATED TRAFFIC SIGNAL PERFORMANCE MEASURES," Mar. 2019.
- [12] T. R. Board, of Sciences, Engineering, and Medicine, *Highway Capacity Manual 7th Edition: A Guide for Multimodal Mobility Analysis*. Washington, DC: The National Academies Press, 2022. doi: 10.17226/26432.
- [13] M. T. Rafique, A. Mustafa, and H. Sajid, "Reinforcement Learning for Adaptive Traffic Signal Control: Turn-Based and Time-Based Approaches to Reduce Congestion," Sep. 2024.
- [14] Y. Zheng, J. Luo, H. Gao, Y. Zhou, and K. Li, "Pri-DDQN: learning adaptive traffic signal control strategy through a hybrid agent," *Complex and Intelligent Systems*, vol. 11, no. 1, Jan. 2025, doi: 10.1007/s40747-024-01651-5.
- [15] A. Agarwal, D. Sahu, R. Mohata, K. Jeengar, A. Nautiyal, and D. K. Saxena, "Dynamic traffic signal control for heterogeneous traffic conditions using Max Pressure and Reinforcement Learning," *Expert Syst Appl*, vol. 254, Nov. 2024, doi: 10.1016/j.eswa.2024.124416.
- [16] F. Ahmad, O. Almarri, Z. Shah, and L. Al-Fagih, "Game theory applications in traffic management: A review of authority-based travel modelling," *Travel Behav Soc*, vol. 32, Jul. 2023, doi: 10.1016/j.tbs.2023.100585.
- [17] Z. Sun *et al.*, "Applications of Game Theory in Vehicular Networks: A Survey," *IEEE Communications Surveys and Tutorials*, vol. 23, no. 4, pp. 2660–2710, 2021, doi: 10.1109/COMST.2021.3108466.
- [18] N. Roupail, A. Tarko, and J. Li, "Traffic Flow at Signalized Intersections," Jan. 2017. Accessed: Apr. 23, 2025. [Online]. Available: <https://www.fhwa.dot.gov/publications/research/operations/tft/chap9.pdf>
- [19] C. Cheng, D. Yao, Y. Zhang, J. Li, and Y. Guo, "A Vehicle Passing Model in Non-signalized Intersections based on Non-cooperative Game Theory," in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, 2019, pp. 2286–2291. doi: 10.1109/ITSC.2019.8917396.
- [20] K. A. Kotb, M. S. Eid, and A. Osama, "Optimizing Adaptive Signalized Intersections: A Cooperative Game Theory Approach," in *Computing in Civil Engineering 2023*, Jan. 2024, pp. 555–563. doi: 10.1061/9780784485224.067.
- [21] M. A. Hossam, Y. Hao, and R. H., "Isolated Traffic Signal Control using Nash Bargaining Optimization," 2017.
- [22] O. D. of T. ODOT, "Turning Movement Counts." Accessed: Sep. 14, 2025. [Online]. Available: <https://odot.ms2soft.com/TDMS.UI/TMC/Analysis/Index?Intersection=9021>